



# KENYA INSTITUTE OF CURRICULUM DEVELOPMENT

STRANDS AND SUBSTRANDS

## AGRICULTURE

GRADE 4, 5 AND 6

**Agriculture key competencies**

1. *Sustainable consumption*
2. *Effective Production*





# STRANDS AND SUBSTRANDS

| Strand                                | Sub strand                                                                                                                                     |  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>1.0 Conserving our Environment</b> | <b>Soil</b> <ul style="list-style-type: none"><li>• Soil particles</li><li>• Uses of soil in farming</li><li>• Compost manure</li></ul>        |  |
|                                       | <b>Water</b> <ul style="list-style-type: none"><li>• Uses of water in farming</li><li>• Water conservation in farming</li></ul>                |  |
|                                       | <b>Living better with wild animals</b>                                                                                                         |  |
|                                       | <b>Growing fruit trees</b>                                                                                                                     |  |
|                                       | <b>Conservation project: Edible crop gardening</b>                                                                                             |  |
| <b>2.0 Domestic animals</b>           | <b>Domestic animals and their uses</b>                                                                                                         |  |
| <b>3.0 Gardening Practices</b>        | <b>Crops for gardening</b> <ul style="list-style-type: none"><li>• Vegetables</li><li>• Cereals</li><li>• Legumes</li></ul>                    |  |
|                                       | <b>Selected gardening practices</b>                                                                                                            |  |
|                                       | <b>Innovative gardening project</b> <ul style="list-style-type: none"><li>• Container gardening</li><li>• Care for container gardens</li></ul> |  |



# Expectations of the Curriculum

## Strand 1: Conserving our Environment

### **Water: Uses of Water**

The uses of water include watering domestic animals and plants. Conduct these activities using the plants and animals available in the locality, or take the learners for an excursion where no animals are available in the school.

### **Water Conservation in Farming**

There are many conservation measures on water as a scarce resource in our day to day life. These have been covered in the previous grades. This section expects conservation of water in farming by utilizing drip irrigation. Drip irrigation in its simplest form utilizes a bottle with water inverted at the base of the plant. In other forms, guide the learners to the next level to utilize a pipe of water, attached to a source of water such as a bucket.



# Expectations of the Curriculum

## Strand 1: Conserving our Environment

### **Living Better with Wild Animals**

Agriculture enterprises may be destroyed by wild animals. The wild animals could destroy animal enterprises as well as crops. This sub strand intends to concentrate on small wild animals that destroy crops and animals with an aim to develop competencies of controlling animals while coexisting with them in the ecosystem. The small animals include but are not limited to the birds, squirrels, monkeys, mongoose and moles among others. This is essential to establish peaceful coexistence of human beings with wild animals in the country, while at the same time conserving the natural environment.



# Expectations of the Curriculum

## Strand 1: Conserving our Environment

### Growing Fruit Trees

This sub strand intends to develop the skills of learners in growing their own fruits and give them opportunity to benefit from their nutrients. Please note that this sub strand concentrates on fruits that grow on hardy trees. Fruit production is a whole process comprising of:

- Collection of fruit seeds
- Fruit seed extraction
- Fruit seed preparation
- Fruit tree nursery bed
- Transplanting, and
- Care for young fruit trees



# Expectations of the Curriculum

## Strand 1: Conserving our Environment

### Conservation Project: Edible Crop Gardening

This sub strand is a progressive concept development of the previous sub strand (growing fruit trees). The essence of this sub strand is that successful production of fruits will only take place if the whole process of taking care of the planted fruit trees is provided. In implementation of this sub strand, the lessons can be planned to run concurrently with another sub strand such that in a week, there is one lesson for ***conservation project: Edible crop garden practices*** while the remaining two lessons are allocated to another sub strand.

This is because taking care of a crop is a process and not a one-off activity.



# **Expectations of the Curriculum**

## **Strand 2: Domestic Animals**

### **Domestic Animals**

Domestic animals are components of agriculture enterprises. This strand aims to develop basic concepts (introductory level) to animals rearing for the purpose of appreciation of animals as a worth part of agriculture while recognizing the age limit of the learner and laying foundation for animal production to be developed at a later grade.



# Expectations of the Curriculum

## Strand 3: Gardening Practices

This strand introduces the concept of gardening which is a vital practice in Kenya today where land, among other agricultural resources, is becoming progressively scarce. Agriculture is a core source of food and nutrition in Kenya. To ensure that we achieve food security, it is important to use any available opportunity to contribute to food production, either for our consumption or for sale. While agricultural space is getting limited, the demand for food is increasing.

Guide the learners to practice the basic crop production of:



# Expectations of the Curriculum

## Strand 3: Gardening Practices

Vegetables such as carrots (root vegetables), spinach (leafy vegetables), and tomatoes (fruit vegetables). However, you may adopt use of other types of vegetables that grow in the locality and that may develop relevant knowledge and skills on vegetables.

Cereals such as wheat, maize and rice. You may also expose them to other cereals to develop relevant knowledge and skills on cereals.

Legumes such as beans, peas and green grams. You may also expose them to other legumes to develop relevant knowledge and skills on legumes.



# **Expectations of the Curriculum**

## **Strand 3: Gardening Practices**

### **Selected Gardening Practices: Direct Sowing of Tiny Seeds**

Direct sowing of seeds is common with large seeds such as those of maize. However, some crops such as carrots have small seeds and require to be planted directly to the seedbed (without the nursery stage). It is with this understanding that the same skills should be learnt by the pupils.

### **Gardening Tools and Equipment**

In this area, guide the learners to identify the tools and equipment that they use in the various gardening practices. It is appropriate to ensure that the learners use appropriate tools and equipment for the various tasks in gardening activities.



# Expectations of the Curriculum

## Strand 3: Gardening Practices

### **Innovative Gardening Project**

The sub strand innovative gardening project aims to ensure that gardening takes an interesting dimension to ensure that every individual learner is empowered to participate in food production regardless of space limitation. Innovative gardening project recognizes that a learner cannot be taught how to be innovative, but can be shown innovative ideas that can trigger their innovativeness.



# Expectations of the Curriculum

## **Container Gardening**

Container gardening implies making space for growing crops in a container.

The size and type of the container varies with the choice of the user, availability of the containers and the crop to be grown.

Container gardens give opportunity to grow crops on ground that could be inappropriate for cropping either because of limited space, spaces with stones or bushes, space between structures, cemented space or on storey buildings

Container gardens provoking innovativeness of the user.

Container gardens empower the learner's self-efficacy by enabling them produce useable crops at their age regardless of landlessness or urban and peri-urban life limitations.



**DO NOT LIMIT AGRIC TO LAND**

**POWER OF Container Gardens**

**Can be used to grow fruits**

**Can be used to grow vegetables**

**Can be used for nursery beds**

**Can be used as plots for cereals**

**CAN BE USED FOR ALL PRACTICES**

**AGRIC WITH MINIMUM LABOUR**



**Sack gardens**

**Plastic container  
gardens**





**Commercial  
germination pots**

**Wooden box  
garden**



## Storey tyre garden



## Fruits and vegetables in Basin gardens





**Hanging container  
garden**



**Trough garden**



**Double tyre garden**



**Deep triple tyre  
garden for fruit trees**



**The Innovations are endless**

Quote: Anonymous

**“I don’t need a farm to produce my  
HEALTHY VEGETABLES, I only need my  
BRAINS”**



**A Hungry Nation  
is an Ugly Nation**

Anonymous